Collaborative Document Editor with AI Writing Assistant

Assignment 1: Requirements Engineering, Architecture & Proof of Concept

Team Size: 3 members
PoC Stack for This Submission: Next.js frontend, FastAPI backend, local SQLite database, LM Studio local LLM API with mock fallback for testing

Current PoC Status for This Submission

This report contains two kinds of content:

1. **Target-system requirements and architecture** for the collaborative editor the team plans to build over the semester.
2. **Current proof-of-concept status** showing what is already implemented in code for Assignment 1.

Implemented in the current submission: - working Next.js dashboard and editor pages - frontend-to-backend document create, load, list, update, delete, export, and version flows - local demo users and backend-enforced document permissions - AI invocation and history flows through `/api/ai/invoke` and `/api/ai/history` - basic WebSocket collaboration room with structured presence and document-update events - role-aware UI states in the frontend with local role switching for demo coverage

Planned for later milestones rather than fully implemented now: - full conflict-safe collaborative editing - production-grade authentication flows - comprehensive frontend automated tests

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Part 1: Requirements Engineering (30%)

1.1 Stakeholder Analysis

Stakeholder	Goals	Concerns	Influence on Requirements
Document Owner / Primary Author	Create, edit, share, and recover documents safely; use AI without losing control over the final text	Overwrites during collaboration; accidental acceptance of poor AI suggestions; unclear version history	Drives fine-grained roles, explicit accept/reject AI UX, and version history with revert support
Collaborator / Editor	Edit concurrently with others and see changes immediately	Lost work during conflicts or reconnects; AI changing text while they are editing nearby	Drives real-time sync, presence awareness, reconnection handling, and soft-lock behavior for AI operations
Reviewer / Commenter	Review content, leave feedback, and inspect how text evolved	Accidentally getting edit or AI powers; losing context after a rewrite	Drives distinct commenter permissions and visible AI/version history
Instructor / Evaluator	Run the proof of concept locally with minimal setup friction	Complex cloud dependencies, missing credentials, brittle install steps	Drives local-first architecture: SQLite, LM Studio, simple `.env` files, and one-command backend startup
DevOps / Maintainer	Keep the app understandable and easy to run on different machines	Too many moving parts, environment drift, and hidden runtime dependencies	Drives a simple monorepo layout, one FastAPI backend process, local SQLite persistence, and clear startup documentation

1.2 Functional Requirements

Real-Time Collaboration

ID	Description	Trigger	Expected Behavior	Acceptance Criteria
FR-COLLAB-001	The system shall propagate document updates to connected collaborators in near real time	A user edits a shared document	Other active editors receive the change over the collaboration channel	Two browser tabs editing the same document on localhost reflect updates within 150 ms (PoC target)
FR-COLLAB-002	The system shall show collaborator presence in an active session	Two or more users are connected to the same document	Each user sees who else is online in the document session	Opening 3 clients on the same document shows 3 active participants in the session list
FR-COLLAB-003	The system shall tolerate temporary disconnection and allow recovery on reconnect	A client loses network connectivity and later reconnects	The client reconnects to the session and resynchronizes the latest document state	After a forced disconnect, reconnecting restores the latest saved state without crashing the session
FR-COLLAB-004	The system shall notify collaborators when a document is reverted to a saved version	Owner triggers a version revert	Active clients receive a visible notification and updated content	Reverting a version updates all connected clients within 3 seconds

AI Writing Assistant

ID	Description	Trigger	Expected Behavior	Acceptance Criteria
FR-AI-001	The system shall allow editors and owners to invoke AI actions on selected text	User selects text and chooses rewrite, summarize, translate, or restructure	The backend submits the request to LM Studio with the selected text and context	A valid request to `/api/ai/invoke` returns a generated suggestion with HTTP 200
FR-AI-002	The system shall present AI output as a suggestion	AI response is received	The suggestion is shown in a	The original document remains unchanged until

ID	Description	Trigger	Expected Behavior	Acceptance Criteria
	rather than silently modifying the document		review area for user approval	the user explicitly accepts the suggestion
FR-AI-003	The system shall log AI interactions for later review	An AI request completes	The request and response are stored in the local database	`/api/ai/history` returns the completed interaction with feature type and timestamp
FR-AI-004	The system shall degrade gracefully if LM Studio is unavailable	An AI request is made while the local LLM server is offline	The system returns a clear error without affecting document editing	The API returns a 502-style failure response and the UI keeps editing enabled
FR-AI-005	The system shall identify the document region currently being processed by AI	AI request starts on selected text	The frontend can mark the affected region as pending	The response payload includes enough context for the frontend to map the suggestion back to the selected region

Document Management

ID	Description	Trigger	Expected Behavior	Acceptance Criteria
FR-DOC-001	The system shall support document create, read, update, and delete operations	User performs a document action	The change is persisted locally in SQLite	CRUD operations on `/api/documents` succeed against the local database
FR-DOC-002	The system shall support named document versions	User saves a version	A version snapshot is created and linked to the document	`/api/documents/{id}/versions` returns the saved version entry

ID	Description	Trigger	Expected Behavior	Acceptance Criteria
FR-DOC-003	The system shall allow reverting a document to a previous version	Owner chooses a version to restore	The document content is replaced by the selected version content	After revert, fetching the document returns the restored content
FR-DOC-004	The system shall expose document metadata needed by the frontend	Frontend loads a document list or detail page	The API returns document id, title, content, and timestamps	Document responses contain id, title, content, `created_at`, and `updated_at`

User Management

ID	Description	Trigger	Expected Behavior	Acceptance Criteria
FR-USER-001	The proof of concept shall support local user identities for testing	Instructor or team member runs the app locally	The system can associate actions with a local test user identity	Demo accounts or seeded users can be used without any external identity provider
FR-USER-002	The system shall enforce role-based access control at the API layer	A user attempts an action beyond their permission	The API rejects the action with a clear authorization error	A commenter attempting an AI invocation receives HTTP 403
FR-USER-003	The system shall distinguish owner, editor, commenter, and viewer roles	A document is shared or accessed	Allowed actions depend on the assigned role	A viewer can read but cannot update or invoke AI
FR-USER-004	The system shall keep authorization rules simple enough for local testing	The app is set up on a new machine	No external auth service is required to test the PoC	The instructor can run the app with only local configuration files

1.3 Non-Functional Requirements

Latency

ID	Requirement	Justification
NFR-LAT-001	Keystroke propagation between two local collaborators shall be ≤ 150 ms end-to-end (P95) in the PoC	On a single local machine, the system should feel effectively real time
NFR-LAT-002	The AI assistant shall acknowledge a request or begin rendering a loading state within 1 second	Users need immediate feedback that the local model request was accepted
NFR-LAT-003	A document up to 1 MB shall load from SQLite and render in the UI within 2 seconds on a normal laptop	The local PoC should feel responsive without requiring tuned infrastructure

Scalability

ID	Requirement	Justification
NFR-SCALE-001	The architecture shall support at least 10 concurrent editors on a single document in the semester project target	This is sufficient for realistic team editing scenarios in the course context
NFR-SCALE-002	The PoC shall support at least 50 active documents in the local database without schema changes	A local evaluator should be able to create enough sample data without performance collapse
NFR-SCALE-003	The backend design shall allow future migration from SQLite to PostgreSQL without rewriting route handlers	This keeps the local-first PoC simple while preserving a path to a larger deployment later

Availability

ID	Requirement	Justification
NFR-AVAIL-001	Core document CRUD must remain usable even if the AI subsystem fails	Editing is the core product value; AI is an enhancement

ID	Requirement	Justification
NFR-AVAIL-002	If the WebSocket collaboration channel disconnects, the user shall be informed within 3 seconds	Silent collaboration failure is confusing and risks duplicate or stale edits
NFR-AVAIL-003	Local data shall persist across backend restarts through the SQLite database file	Instructors need to restart the app without losing demo data

Security & Privacy

ID	Requirement	Justification
NFR-SEC-001	For local development, only the selected text and small surrounding context shall be sent to LM Studio	Even though LM Studio runs locally, data minimization remains good practice
NFR-SEC-002	Secrets and machine-specific paths shall be stored in `.env` files and excluded from version control	This keeps the repository portable and avoids accidental credential leaks
NFR-SEC-003	The PoC shall avoid mandatory third-party cloud providers for authentication, storage, or AI inference	The assignment specifically benefits from a setup that works offline or on a single machine

Usability

ID	Requirement	Justification
NFR-USE-001	The UI shall clearly separate authored text from AI suggestions	Users must understand what is human-written and what is machine-suggested
NFR-USE-002	The UI shall remain understandable when multiple collaborators are active by collapsing excess presence indicators	Presence is useful, but too many indicators create clutter
NFR-USE-003	The core flows shall remain keyboard-accessible	Accessibility matters even in a student proof of concept

1.4 User Stories and Scenarios

Collaboration Scenarios

US-01: As an editor, I want my changes to appear quickly for another editor so that collaboration feels immediate.
- Acceptance criteria: Two localhost browser windows editing the same document show updates within 150 ms in the PoC.

US-02: As an editor, I want to reconnect to an editing session after a connection drop so that I can continue without reopening the whole app.
- Acceptance criteria: Reconnecting restores the latest server state and rejoins the presence list.

US-03: As a document owner, I want to revert to a saved version so that I can undo a bad change safely.
- Acceptance criteria: Revert restores the saved content and creates a visible update for connected collaborators.

AI Assistant Workflows

US-04: As an editor, I want to summarize a selected paragraph so that I can condense long content quickly.
- Acceptance criteria: Sending selected text to the AI endpoint returns a summary suggestion that does not overwrite the document automatically.

US-05: As an editor, I want to translate a selected section so that I can prepare multilingual content.
- Acceptance criteria: The AI response is returned in the requested target language.

US-06: As an editor, I want to partially apply an AI rewrite so that I stay in control of the final text.
- Acceptance criteria: The suggestion is presented separately and the user can manually edit or selectively apply it in the frontend.

US-07: As a collaborator, I want to know when another user is running AI on a passage so that I avoid editing the same region at the same moment.
- Acceptance criteria: The frontend can show a pending marker for the targeted region.

Document Lifecycle

US-08: As an owner, I want to create and rename documents so that I can organize my work.
- Acceptance criteria: The document list updates after create and rename actions.

US-09: As an owner, I want to browse saved versions of a document so that I can understand how it evolved over time.
- Acceptance criteria: The version list includes labels and timestamps.

Access Control and Roles

US-10: As a commenter, I want the system to explain why I cannot invoke AI so that the restriction feels intentional rather than broken.
- Acceptance criteria: The user receives a clear permission message instead of a silent failure.

US-11: As a viewer, I want editing to be blocked gracefully so that I understand my current level of access.
- Acceptance criteria: The editor enters read-only mode and shows a concise explanation.

US-12: As an instructor, I want to run the system without cloud signup steps so that testing the submission is quick and reproducible.
- Acceptance criteria: Following the README with local `.env` files is sufficient to run the PoC.

1.5 Requirements Traceability Matrix

User Story	Functional Requirements	Architecture Components
US-01	FR-COLLAB-001, FR-COLLAB-002	Next.js Frontend, FastAPI Collaboration Gateway
US-02	FR-COLLAB-003	FastAPI Collaboration Gateway, SQLite Document Store
US-03	FR-COLLAB-004, FR-DOC-002, FR-DOC-003	FastAPI Document Service, SQLite
US-04	FR-AI-001, FR-AI-002, FR-AI-003	Next.js Frontend, FastAPI AI Service, LM Studio Connector
US-05	FR-AI-001, FR-AI-002	FastAPI AI Service, LM Studio Connector
US-06	FR-AI-002, FR-AI-005	Next.js Frontend, FastAPI AI Service

User Story	Functional Requirements	Architecture Components
US-07	FR-AI-005, FR-COLLAB-002	Next.js Frontend, FastAPI Collaboration Gateway
US-08	FR-DOC-001, FR-DOC-004	FastAPI Document Service, SQLite
US-09	FR-DOC-002, FR-DOC-003	FastAPI Document Service, SQLite
US-10	FR-USER-002, FR-USER-003	FastAPI Auth/Permission Layer, Next.js Frontend
US-11	FR-USER-002, FR-USER-003	FastAPI Auth/Permission Layer, Next.js Frontend
US-12	FR-USER-004, NFR-SEC-003	Submission README, FastAPI Backend, SQLite, LM Studio

Every user story maps to at least one functional requirement. Every functional requirement maps to at least one architectural component in Part 2.

Part 2: System Architecture (45%)

2.1 Architectural Drivers

Ranked by influence on the design:

1. **Low-friction local setup**
   The instructor should be able to run the system on one machine with local services only. This directly drove the choice of SQLite, LM Studio, FastAPI, and a simple repo structure.

2. **Real-time collaboration quality**
   The product promise is collaborative editing. This drove the decision to keep a WebSocket-based collaboration channel in the backend architecture.

3. **AI usefulness with acceptable latency**
   AI must feel integrated, not bolted on. This drove selected-text prompts, explicit suggestion review, and direct LM Studio integration through a single backend service.

4. **Privacy and predictability**
   Running the model locally reduces data exposure and avoids third-party API instability. This shaped the AI integration design and security assumptions.

5. **Team delivery speed**
   A 3-person team needs clear boundaries. This drove the split between frontend, backend/data, and AI/integration responsibilities inside a single submission repository.

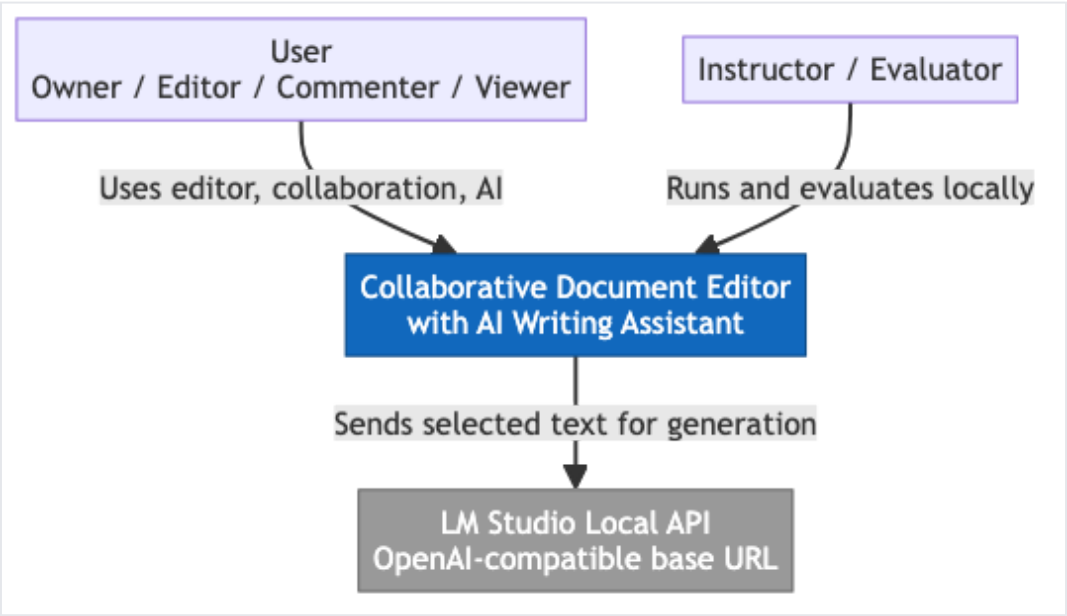
2.2 System Design Using the C4 Model

Level 1 — System Context Diagram

```
graph TB
    User["User<br/>Owner / Editor / Commenter / Viewer"]
    Instructor["Instructor / Evaluator"]
    System["Collaborative Document Editor<br/>with AI Writing Assistant"]
    LM["LM Studio Local API<br/>OpenAI-compatible base URL"]

    User -->|"Uses editor, collaboration, AI"| System
    Instructor -->|"Runs and evaluates locally"| System
    System -->|"Sends selected text for generation"| LM

    style System fill:#1168bd,stroke:#0b4884,color:#fff
    style LM fill:#999,stroke:#666,color:#fff
```



Explanation: The system depends on only one external runtime component for the PoC: LM Studio running locally. No external database service, cloud storage, or third-party identity provider is required for evaluation.

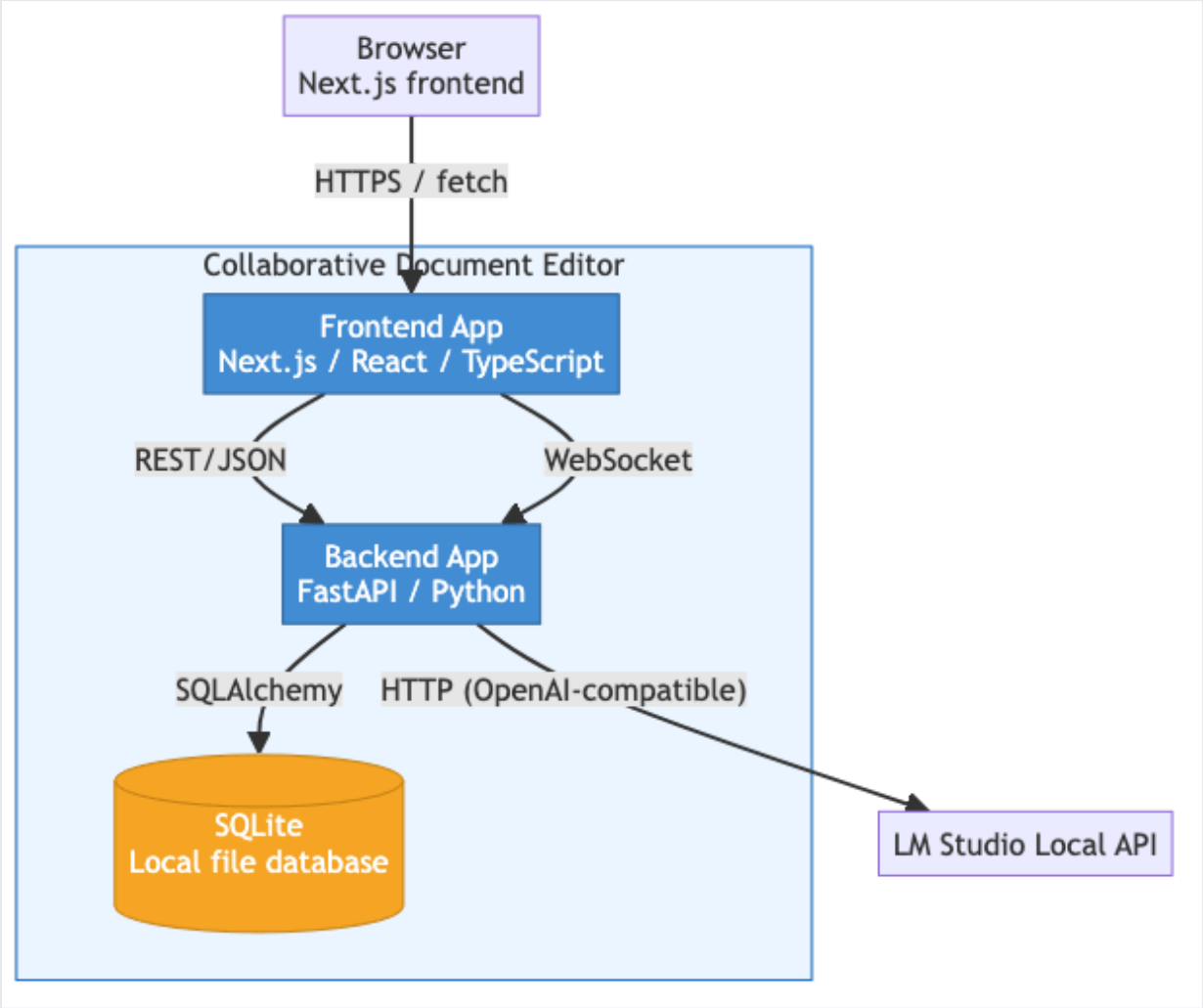
Level 2 — Container Diagram

```
graph TB
    Browser["Browser<br/>Next.js frontend"]
    LM["LM Studio Local API"]

    subgraph system ["Collaborative Document Editor"]
        Frontend["Frontend App<br/>Next.js / React / TypeScript"]
        Backend["Backend App<br/>FastAPI / Python"]
        DB[("SQLite<br/>Local file database")]
    end

    Browser -->|"HTTPS / fetch"| Frontend
    Frontend -->|"REST/JSON"| Backend
    Frontend -->|"WebSocket"| Backend
    Backend -->|"SQLAlchemy"| DB
    Backend -->|"HTTP (OpenAI-compatible)"| LM

    style system fill:#eef6ff,stroke:#1168bd
    style Frontend fill:#438dd5,stroke:#2e6295,color:#fff
    style Backend fill:#438dd5,stroke:#2e6295,color:#fff
    style DB fill:#f5a623,stroke:#c7841a,color:#fff
```



Container responsibilities:

Container	Responsibility	Technology
Frontend App	Document pages, editor shell, AI panel, presence UI, and permission-aware interactions	Next.js, React, TypeScript
Backend App	REST API, WebSocket collaboration endpoint, document/version persistence, AI proxy to LM Studio, permission checks	FastAPI, SQLAlchemy, Python
SQLite	Local persistence for users, documents, versions, permissions, and AI history	SQLite
LM Studio Local API	Local LLM inference for rewrite, summarize, translate, and restructure features	LM Studio server

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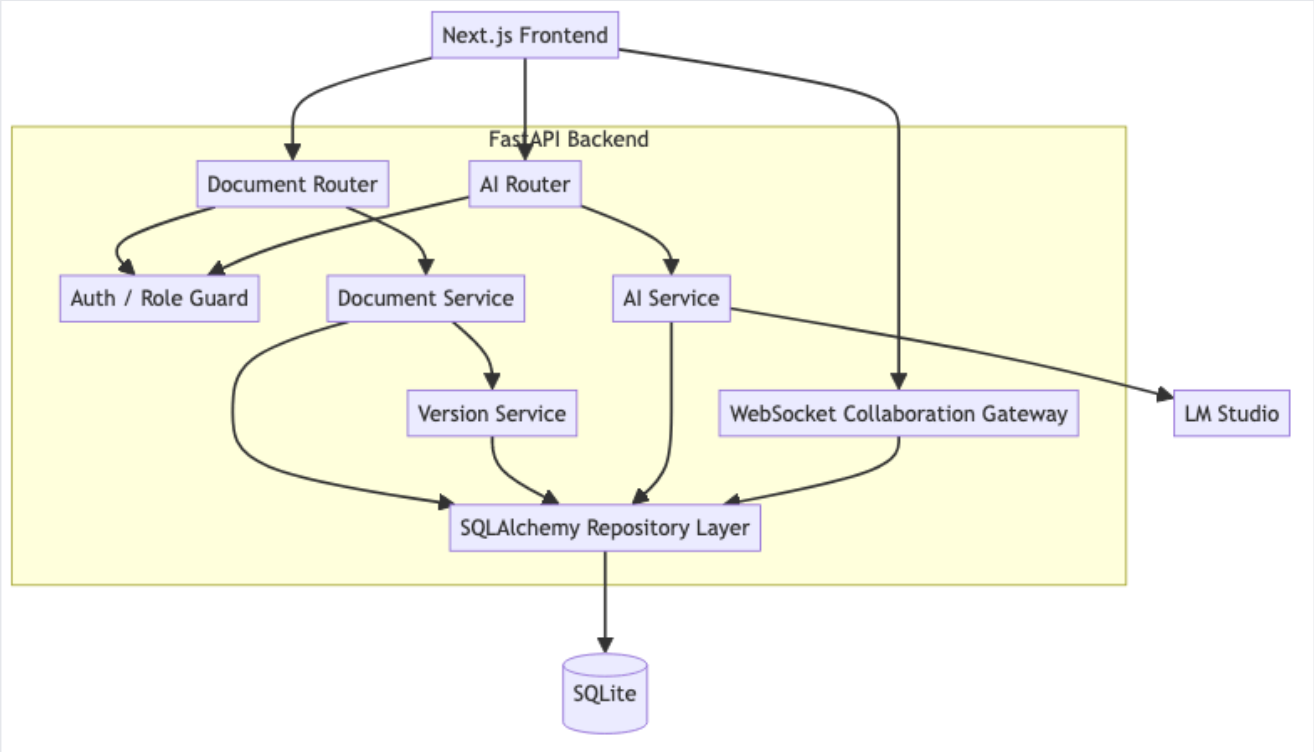
Level 3 — Component Diagram (FastAPI Backend)

```
graph TB
    Frontend["Next.js Frontend"]
    LM["LM Studio"]
    DB[("SQLite")]

    subgraph backend ["FastAPI Backend"]
        DocRouter["Document Router"]
        AiRouter["AI Router"]
        WsGateway["WebSocket Collaboration Gateway"]
        AuthLayer["Auth / Role Guard"]
        DocService["Document Service"]
        VersionService["Version Service"]
        AiService["AI Service"]
        Repo["SQLAlchemy Repository Layer"]
    end

    Frontend --> DocRouter
    Frontend --> AiRouter
    Frontend --> WsGateway

    DocRouter --> AuthLayer
    AiRouter --> AuthLayer
    DocRouter --> DocService
    DocService --> VersionService
    DocService --> Repo
    VersionService --> Repo
    AiRouter --> AiService
    AiService --> Repo
    AiService --> LM
    WsGateway --> Repo
    Repo --> DB
```



Component responsibilities:

Component	What It Does	Dependencies	Interface
Document Router	Handles CRUD and version endpoints	Auth Layer, Document Service	`/api/documents/*`
AI Router	Validates AI requests and returns suggestions/history	Auth Layer, AI Service	`/api/ai/*`
WebSocket Collaboration Gateway	Maintains live document sessions and broadcasts updates	Repository Layer	`/ws/documents/{documentId}`
Auth / Role Guard	Enforces owner/editor/commenter/viewer permissions	User/permission data	Middleware / dependencies
Document Service	Business rules for create, update, delete, list, and revert	Repository Layer	Internal
Version Service	Creates version snapshots and handles reverts	Repository Layer	Internal
AI Service	Builds prompts, calls LM Studio, logs AI interactions	LM Studio, Repository Layer	Internal

Component	What It Does	Dependencies	Interface
Repository Layer	Reads and writes SQLite data through SQLAlchemy	SQLite	Internal

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Feature Decomposition

1. Rich-Text Editor & Frontend State

- What it does: Hosts the editor UI, document pages, AI review panel, and presence indicators.
- Depends on: FastAPI REST endpoints and WebSocket collaboration endpoint.
- Interface: Fetch calls to `/api/*` and WebSocket sessions to `/ws/documents/{id}`.

2. Real-Time Synchronization Layer

- What it does: Manages active document sessions and broadcasts updates to connected clients.
- Depends on: FastAPI WebSocket support and persisted document state in SQLite.
- Interface: WebSocket messages carrying document updates and presence events.

3. AI Assistant Module

- What it does: Builds prompts from selected text, sends them to LM Studio, returns suggestions, and records history.
- Depends on: LM Studio local API and local persistence for interaction logs.
- Interface: `/api/ai/invoke` and `/api/ai/history`.

4. Document Storage & Versioning

- What it does: Stores document content, metadata, and version snapshots.
- Depends on: SQLite through SQLAlchemy.
- Interface: `/api/documents`, `/api/documents/{id}`, `/api/documents/{id}/versions`.

5. Authentication & Authorization

- What it does: Associates actions with local user identities and enforces per-role permissions.
- Depends on: User and permission tables.

- Interface: Route dependencies and future `/api/auth/*` endpoints.

6. API Layer

- What it does: Exposes a stable contract between the Next.js frontend and the backend.
- Depends on: All backend services.
- Interface: REST endpoints and a WebSocket endpoint.

AI Integration Design

Context and Scope

The AI service sends: - the selected text, - optionally a small surrounding context window, - the requested feature type.

The full document is not sent by default. This keeps prompts focused, reduces local inference latency, and minimizes accidental data exposure even though the LLM is running locally.

Suggestion UX

The AI output is shown in a side panel or suggestion area rather than applied inline immediately. - The original text stays unchanged until the user accepts it. - The user can edit the suggestion before applying it. - Partial application is handled in the frontend by letting the user copy or apply only part of the response.

AI During Collaboration

- The invoking user marks the selected region as “AI pending”.
- Other collaborators can still edit outside that region.
- If the region changes before the response is applied, the frontend should warn the user that the suggestion may now be stale.

This is a pragmatic compromise for the semester project: predictable and easy to explain, without trying to auto-merge AI edits into a moving target.

Prompt Design

Prompt construction is template-based in backend code. - Each feature type maps to a specific instruction pattern. - The prompt builder inserts selected text and surrounding context. - Later iterations can move prompt templates into dedicated files without changing the route contract.

Model and Cost Strategy

Because LM Studio runs locally, the main constraint is not per-token billing but local compute capacity. - The PoC uses a single configured local model. - The backend can later route simple tasks to smaller local models and harder tasks to larger ones. - If the model server is busy or unavailable, the backend returns a clear failure and leaves document editing unaffected.

API Design

Document CRUD

Method	Endpoint	Request Body	Response	Errors
GET	`/api/health`	—	`200 { status, app_name }`	—
GET	`/api/documents`	—	`200 [{ id, title, content, created_at, updated_at }]`	500
POST	`/api/documents`	`{ title, content, save_initial_version }`	`201 { id, title, content, created_at, updated_at }`	400
GET	`/api/documents/:id`	—	`200 { id, title, content, created_at, updated_at }`	404
PATCH	`/api/documents/:id`	`{ title?, content?, create_version?, version_label? }`	`200 { ...document }`	404
DELETE	`/api/documents/:id`	—	`204`	404

Method	Endpoint	Request Body	Response	Errors
GET	`/api/documents/:id/versions`	—	`200 [{ id, document_id, label, content, created_at }]`	404

Real-Time Session Management

Protocol	Endpoint	Description
WebSocket	`/ws/documents/:id`	Broadcasts presence and update messages among clients connected to the same document

Message examples: - `{"type":"join","user":"alice@example.com"}` - `{"type":"update","content":"...delta or document payload..." }`

AI Assistant

Method	Endpoint	Request Body	Response	Errors
POST	`/api/ai/invoke`	`{ feature, selected_text, surrounding_context?, target_language?, document_id? }`	`200 { feature, output_text, model_name, provider, status, mocked }`	403, 404, 502
GET	`/api/ai/history`	`document_id?`, `feature?`, `limit?`	`200 [{ id, document_id, user_id, feature, prompt_excerpt, response_text, model_name, status, created_at }]`	404, 500

User & Permission Management

For the PoC, the authorization model is deliberately simple: - local demo users, - per-document roles, - backend-enforced permission checks.

Current implementation note: - the backend already exposes local demo users through header-based identity selection - document permissions are enforced on backend routes and reflected in the frontend workspace - the current PoC does not include password login, OAuth, or long-lived auth sessions

Suggested future endpoints: - `POST /api/auth/login` - `POST /api/auth/logout` - `POST /api/auth/refresh`

API Style Justification

- **REST** for CRUD and AI invocation because it is simple, testable, and well supported by FastAPI.
- **WebSocket** for collaboration because real-time push is necessary for editing sessions.
- A separate AI microservice is intentionally avoided in the PoC to keep setup friction low.

Long-Running AI Operations

In this PoC design, the frontend sends a regular HTTP request to `/api/ai/invoke` and shows a loading state until the local model returns. This is acceptable for the current implementation because LM Studio runs on the same machine and the goal is a low-complexity local proof of concept.

If inference latency becomes a problem later, the same backend can be extended with: - SSE for token streaming, or - a job-based status endpoint.

Error Handling Strategy

Standard error shape:

```
{
  "detail": "LM Studio request failed. Check the local server and model set
}
```

Client-visible interpretation:

HTTP Status	Meaning	Client Action
200	Success	Show the suggestion
403	Permission denied	Explain the required role
404	Missing document	Redirect or show not found

HTTP Status	Meaning	Client Action
502	LM Studio unavailable or invalid response	Keep editing enabled and show an AI-specific error

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Authentication & Authorization

Why Authentication Is Needed

Documents and AI history belong to specific users and roles. Even in a local PoC, the system must differentiate between owner, editor, commenter, and viewer behavior.

Role Definitions

Role	Read	Edit	Comment	Invoke AI	Revert Versions	Share
Owner	✅	✅	✅	✅	✅	✅
Editor	✅	✅	✅	✅	❌	❌
Commenter	✅	❌	✅	❌	❌	❌
Viewer	✅	❌	❌	❌	❌	❌

Key choices: - AI invocation is limited to editors and owners because it proposes content changes. - Version revert is owner-only because it affects all collaborators. - The PoC uses local identities instead of external OAuth to reduce setup friction.

Privacy Considerations

- The LLM runs locally through LM Studio, so document text does not need to leave the evaluator's machine.
- Only the selected text and limited context are sent to the model.
- AI interaction history is stored in the same local database as the rest of the app data.

Communication Model

Push-Based Collaboration

The system uses a push-based WebSocket model for collaboration. - On document open, the frontend loads the current saved state through REST. - It then opens a WebSocket connection for presence and live updates. - If the socket disconnects, the UI shows a reconnecting indicator and attempts to reconnect.

Why Not Polling?

Polling is simpler but unsuitable for collaboration because: - it adds avoidable latency, - it wastes requests, - it creates a worse user experience when two people are typing actively.

Open and Reconnect Flow

1. Frontend fetches the document from the REST API.
2. Frontend opens `ws://.../ws/documents/{id}`.
3. Backend registers the client in the document room.
4. Clients broadcast updates and presence changes.
5. On disconnect, the client reconnects and reloads the latest server state if needed.

2.3 Code Structure & Repository Organization

Monorepo vs. Multi-Repo

Decision: single submission repository with two app folders.

This is a simple single-repository submission. The goal is clarity, not tooling sophistication. The repository root contains both application folders and the assignment documents, so evaluators can see the code and the written deliverables in one place.

Directory Layout

```
AI1220B-task-1/
├── README.md
├── report.pdf
├── report.md
├── meeting_log.md
├── diagrams/
├── team-task-division.md
├── frontend/
│   ├── README.md
```

```
│   ├── app/
│   │   ├── components/
│   │   ├── documents/[id]/
│   │   ├── lib/
│   │   ├── globals.css
│   │   ├── layout.tsx
│   │   └── page.tsx
│   ├── public/
│   ├── package.json
│   └── .env.example
└── backend/
    ├── README.md
    ├── app/
    │   ├── main.py
    │   ├── config.py
    │   ├── database.py
    │   ├── models.py
    │   ├── schemas.py
    │   ├── services.py
    │   ├── realtime.py
    │   └── routers/
    ├── data/
    ├── requirements.txt
    └── .env.example
```

Shared Code

To keep the PoC simple, there is no separate shared package yet. - The backend OpenAPI schema acts as the source of truth for the API contract. - The frontend can either define lightweight matching TypeScript types or later generate them from FastAPI's OpenAPI output.

This avoids premature repository complexity while still giving a clean upgrade path.

Configuration Management

- Backend environment variables live in `backend/.env`.
- Frontend environment variables live in `frontend/.env.local`.
- The repository tracks only `.env.example` files.
- Important local settings: SQLite path, LM Studio base URL, LM Studio model name, frontend API base URL.

Testing Structure

- **Backend tests:** route and service tests under `backend/tests/` covering documents, permissions, versions, AI flows, and collaboration behavior.
- **Frontend automated tests:** not included yet in the current submission.
- **Integration coverage:** backend tests use a temporary SQLite database and mocked LM Studio when needed.
- **End-to-end coverage:** the final PoC walkthrough is validated through manual frontend-backend demo flows.

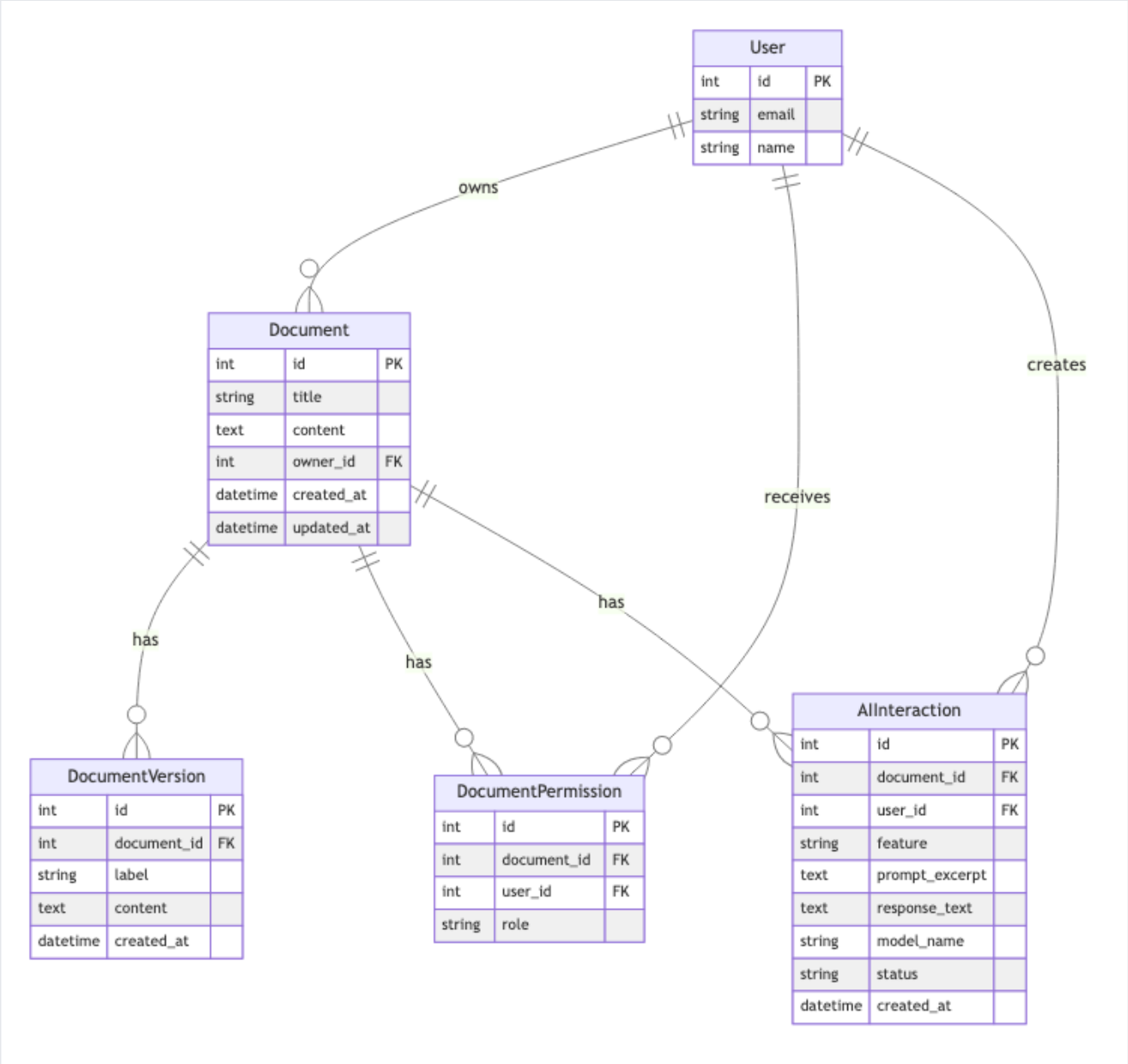
For repeatable automated tests, the backend supports a mock AI mode instead of requiring the real local model every time.

2.4 Data Model

Entity-Relationship Diagram

```
erDiagram
    User {
        int id PK
        string email
        string name
        string global_role
    }

    Document {
        int id PK
        string title
        text content
        int owner_id FK
        datetime created_at
        datetime updated_at
    }

    DocumentVersion {
        int id PK
        int document_id FK
        string label
        text content
        datetime created_at
    }
```

```
    DocumentPermission {
        int id PK
        int document_id FK
        int user_id FK
        string role
    }

    AIInteraction {
        int id PK
        int document_id FK
        int user_id FK
        string feature
        text prompt_excerpt
        text response_text
        string model_name
        string status
        datetime created_at
    }

    User ||--o{ Document : owns
    User ||--o{ DocumentPermission : receives
    User ||--o{ AIInteraction : creates
    Document ||--o{ DocumentVersion : has
    Document ||--o{ DocumentPermission : has
    Document ||--o{ AIInteraction : has
```



Data Model Decisions

Document representation

The PoC stores document content as text in SQLite. This keeps the current implementation easy to inspect and debug. If the collaboration layer later adopts CRDT snapshots, the content field can evolve to structured JSON without breaking the high-level model.

Versioning

Each version stores a content snapshot and optional label. Reverting does not delete history; it restores previous content through a new user action.

AI interaction history

Each interaction stores: - the related document, - feature type, - a prompt excerpt, - the generated response, - model name, - status, - timestamp.

This is enough for auditing and UI history panels later.

Permissions and sharing
The model supports per-document roles. Link sharing and team-based sharing are future extensions, but the role structure already leaves room for them.

2.5 Architecture Decision Records (ADRs)

ADR-001: Use Next.js for the Frontend and FastAPI for the Backend

Field	Content
Status	Accepted
Context	The team needs a modern frontend and a backend that is fast to scaffold, easy to document, and convenient for local testing.
Decision	Use Next.js for the frontend and a single FastAPI backend for REST, WebSocket, and AI integration.
Consequences	Positive: fast setup, built-in API docs, simple Python backend structure, easy local development. Negative: fewer shared-language types than an all-TypeScript stack.
Alternatives	Node/Express was rejected for this submission because FastAPI offers quicker scaffolding and clearer local API documentation.

ADR-002: Use SQLite for the Proof of Concept Database

Field	Content
Status	Accepted
Context	The instructor should be able to run the project locally without separately installing and configuring a database server.
Decision	Use a local SQLite file for the PoC.
Consequences	Positive: zero external DB setup, portable, easy to inspect. Negative: limited concurrency and weaker fit for production-scale collaboration.

Field	Content
Alternatives	PostgreSQL was considered, but rejected for the PoC because it adds avoidable setup friction.

ADR-003: Use LM Studio Through Its OpenAI-Compatible Local API

Field	Content
Status	Accepted
Context	The user plans to use LM Studio and already has a base URL for it. The assignment also benefits from avoiding third-party cloud AI dependencies.
Decision	The backend calls LM Studio over its OpenAI-compatible HTTP API.
Consequences	Positive: local privacy, no external API key dependency for grading, stable local workflow. Negative: inference speed depends heavily on the evaluator's hardware and chosen model.
Alternatives	Cloud-hosted LLM APIs were rejected for the PoC because they add credentials, billing, and network dependency.

ADR-004: Keep the PoC as a Simple Monorepo-Style Submission

Field	Content
Status	Accepted
Context	A 3-person team needs a shared codebase with low coordination overhead.
Decision	Keep both apps in one repository root under `frontend/` and `backend/`, alongside the assignment documents.
Consequences	Positive: easy to understand, easy to submit, fewer moving pieces. Negative: no dedicated shared package yet, so frontend/backend contracts must be kept aligned carefully.
Alternatives	Multi-repo and heavier workspace tooling were rejected because they add complexity without helping the PoC.

Part 3: Project Management & Team Collaboration (15%)

3.1 Team Structure & Ownership

Role Assignments (3-person team)

Member	Primary Ownership	Scope
Member A	Frontend & Editor UX	Next.js pages, dashboard, editor shell, AI panel, presence UI, and API integration from the frontend
Member B	Backend & Data	FastAPI routes, SQLite schema, document/version logic, permission checks, API documentation
Member C	AI & Collaboration Integration	LM Studio integration, prompt design, AI history, WebSocket collaboration endpoint, local setup polish

Cross-Cutting Feature Handling

For features that touch all three areas: 1. Define the API contract first. 2. Build against a simple mock or stub. 3. Integrate in short vertical slices.

The most cross-cutting feature in this project is AI-assisted editing because it spans UI, backend logic, and local model integration.

Decision-Making Process

- Decisions inside an ownership area are made by the owner.
- Cross-cutting changes are discussed briefly by the full team.
- Important choices are documented as short ADRs.

3.2 Development Workflow

Branching Strategy

- Use short-lived feature branches.
- Merge to `main` only through pull requests.
- Keep branches small enough to review quickly.

Suggested naming: - `frontend/editor-shell` - `backend/document-crud` - `ai/lm-studio-integration`

Code Review

- Every PR requires at least 1 reviewer.
- PRs touching public API contracts should be reviewed by both frontend and backend owners.
- Review focus: correctness, clarity, and consistency with the report.

Issue Tracking & Task Assignment

- Use a simple Kanban board: `Backlog`, `In Progress`, `Review`, `Done`.
- Split issues by vertical features and shared infrastructure.
- Tag issues with `frontend`, `backend`, `ai`, `collaboration`, `docs`, or `testing`.

Communication

- Daily async updates in chat.
- One short weekly sync for planning and integration.
- Architectural decisions go into markdown, not only chat history.

3.3 Development Methodology

Choice: lightweight sprint-based workflow.

This team is small, so the process should stay minimal: - 1-week planning horizon, - async updates, - short review/demo at the end of the week.

Prioritization

- **Must have:** local startup, document CRUD, versioning, AI invocation, collaboration starter.
- **Should have:** role enforcement, polished AI review UI, stable reconnect behavior.
- **Could have:** tracked AI diffs, link sharing, export.

Non-Visible Work

Local setup, test scaffolding, and README quality are explicitly planned tasks because they directly affect grading and reproducibility.

3.4 Risk Assessment

Risk 1: LM Studio Performance Varies by Machine

Aspect	Detail
Description	AI latency and quality depend on the evaluator's local hardware and chosen model.
Likelihood	High
Impact	Medium
Mitigation	Keep prompts short, use selected text only, document recommended model settings, and support `LLM_MOCK=true` for testing.
Contingency	Demo the AI flow with a smaller local model or mock mode if hardware is limited.

Risk 2: Collaboration Logic Becomes More Complex Than the Starter Suggests

Aspect	Detail
Description	A simple broadcast WebSocket template is not the same as full conflict-safe collaborative editing.
Likelihood	High
Impact	High
Mitigation	Treat the current WebSocket layer as a scaffold and plan a dedicated iteration for CRDT integration.
Contingency	Narrow the demo scope to basic shared-session updates if full conflict resolution is not ready.

Risk 3: Architecture and Code Drift Apart

Aspect	Detail
Description	The report can become outdated if the implementation changes quickly.
Likelihood	Medium
Impact	High
Mitigation	Update the report whenever the team changes the stack, API contract, or repository structure.
Contingency	Reserve a final alignment pass before submission.

Risk 4: Local Environment Differences Cause Setup Failures

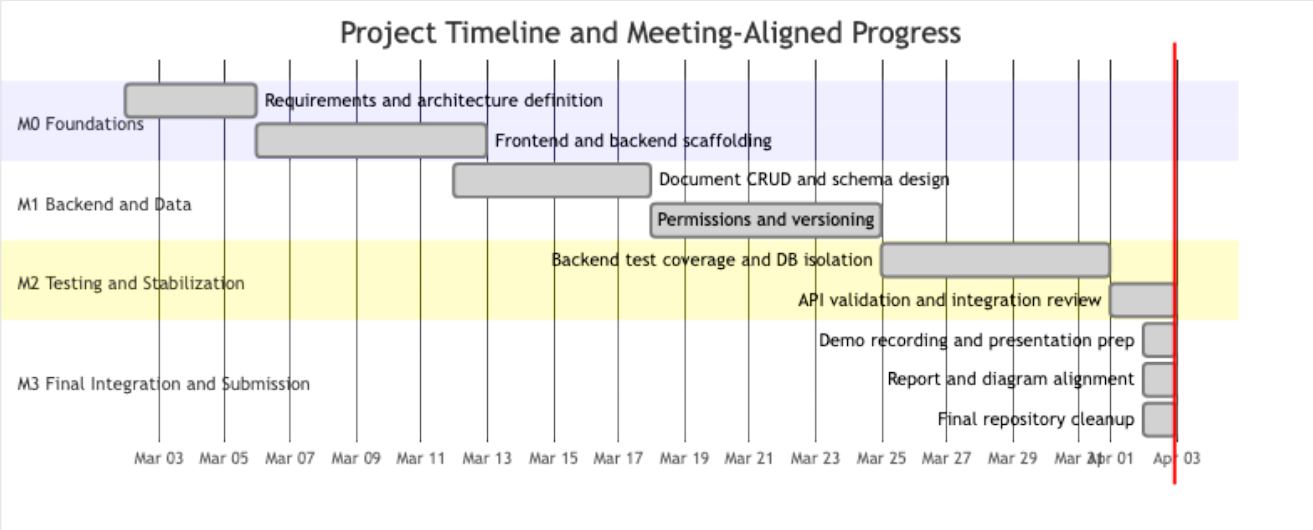
Aspect	Detail
Description	Python version, Node version, or LM Studio configuration may differ between machines.
Likelihood	Medium
Impact	Medium
Mitigation	Keep startup commands simple, document required versions, and avoid unnecessary dependencies.
Contingency	Provide a short troubleshooting section in the submission README.

Risk 5: API Contract Mismatch Between Frontend and Backend

Aspect	Detail
Description	The frontend may assume payload shapes that no longer match the backend.
Likelihood	Medium
Impact	Medium
Mitigation	Treat FastAPI's OpenAPI schema as the contract source of truth and validate frontend integration against it regularly.
Contingency	Freeze the contract for the milestone and defer non-critical shape changes.

3.5 Timeline and Milestones

```
gantt
    title Project Timeline
    dateFormat YYYY-MM-DD
    axisFormat %b %d

    section M0 Foundations
    Revise report and finalize stack          :done, m0a, 2026-03-26, 6d
    Scaffold frontend and backend             :done, m0b, 2026-03-29, 4d

    section M1 Local PoC
    Document CRUD in FastAPI                  :m1a, 2026-04-02, 5d
    Frontend document pages                   :m1b, 2026-04-02, 5d
    LM Studio integration baseline            :m1c, 2026-04-05, 4d

    section M2 Collaboration
    WebSocket room handling                   :m2a, 2026-04-09, 5d
    Presence indicators in UI                 :m2b, 2026-04-09, 5d
    Reconnect handling                        :m2c, 2026-04-12, 4d

    section M3 AI Workflow
    AI suggestion review panel                :m3a, 2026-04-16, 5d
    AI history and logging                    :m3b, 2026-04-16, 4d
    Role checks for AI                        :m3c, 2026-04-18, 3d

    section M4 Polish
    Version revert UX                         :m4a, 2026-04-21, 4d
    Integration testing                       :m4b, 2026-04-21, 5d
    Final docs and demo prep                  :m4c, 2026-04-25, 3d
```



Milestone Acceptance Criteria

Milestone	Target Date	Acceptance Criteria
M0: Foundations	2026-03-05	Completed. Team scope, stack choices, C4 design, and module boundaries were agreed and documented.
M1: Backend and Data	2026-03-18	Completed. Document CRUD, user roles, permissions, version creation, and revert behavior were implemented in the backend.
M2: Testing and Stabilization	2026-04-01	Completed. Backend tests passed, API behavior was validated, and integration issues were reviewed before demo preparation.
M3: Final Integration and Submission	2026-04-02	Completed. Demo flow, presentation materials, report updates, and repository preparation were completed for submission.

Part 4: Proof of Concept (10%)

4.1 Current PoC Scope

The current repository demonstrates the architectural skeleton required by the assignment rather than a feature-complete product.

Implemented in code today: - a working frontend dashboard that lists documents and creates new ones - a document editor page that loads and updates document content through the backend API - AI assistant controls that send selected text to the backend for rewrite, summarize, translate, and restructure requests - version list, snapshot creation, and owner-only revert controls - local demo users, permissions management, and export actions - a collaboration room connection with presence sync, reconnect state, and structured live update events

Intentionally not complete yet: - real authentication screens or login flow - cursor-level collaborative editing or CRDT-based conflict resolution - comprehensive frontend automated tests

4.2 Frontend-to-Backend Communication

The PoC satisfies the core integration requirement:

- `POST /api/documents` is used from the dashboard to create a document
- `GET /api/documents/:id` is used to load the editor
- `PATCH /api/documents/:id` is used to save document changes
- `POST /api/documents/:id/versions` is used to create version snapshots
- `POST /api/documents/:id/versions/:version_id/revert` is used to restore an older version
- `GET /api/documents/:id/permissions` and `POST /api/documents/:id/permissions` are used to manage sharing
- `POST /api/ai/invoke` is used to request AI suggestions
- `GET /api/users` and `GET /api/users/me` are used to load local demo identities
- `WS /ws/documents/:id` is used for live room state and update broadcasts

This proves that the frontend exists, loads, and communicates with the backend using the contract described in Part 2.

4.3 Data Contract Validation

The current frontend code expects and uses the same document shape described in the API section:

```
{
  "id": 1,
  "title": "Demo project brief",
  "content": "Document body text",
  "created_at": "2026-04-01T10:20:27",
```

```
  "updated_at": "2026-04-01T10:20:27"
}
```

The AI request/response flow also matches the current backend contract: - request: `feature`, `selected_text`, `surrounding_context`, optional `target_language`, optional `document_id` - response: `feature`, `output_text`, `model_name`, `provider`, `status`, `mocked`

4.4 Repository and README Fit

The repository is organized around two application folders: - `frontend/` for the Next.js app - `backend/` for the FastAPI app

The README explains: - how to install dependencies - how to configure `.env` files - how to start the backend and frontend - what the PoC demonstrates - what is intentionally not implemented yet

4.5 PoC Conclusion

For Assignment 1, the PoC goal is satisfied: the frontend communicates with the backend, the main data contracts are visible in working code, the repository structure is understandable, and the setup remains local-first and reproducible.